

Explainable Toxicity Prediction Through XAI and Large Language Models

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Abstract—Machine learning models are increasingly used for molecular toxicity prediction, but their predictions are often difficult to interpret. This work presents an explainable toxicity prediction workflow that combines supervised classification, SHAP, and LIME explanations, a consensus strategy to compare their outputs, and a Large Language Model (LLM) used exclusively as a communication layer. Several classifiers were evaluated using physicochemical descriptors, and the three best-performing models were selected for explainability analysis, with Extra Trees achieving the highest F1-score. SHAP and LIME were compared to identify consistent and conflicting feature attributions, while LLM generated natural-language explanations constrained by the XAI evidence. The generated narratives were then evaluated using fidelity, hallucination, and factuality metrics to quantify how accurately they preserved the underlying explanations. The results show that the proposed workflow enables transparent communication of model predictions while revealing that high descriptor coverage does not necessarily imply faithful ranking or fully supported narratives. These findings highlight the importance of quantitatively evaluating LLM-generated explanations before using them to communicate the XAI results.

Index Terms—Explainable Artificial Intelligence, toxicity prediction, Large Language Models, SHAP, LIME

I. INTRODUCTION

Machine Learning (ML) models are increasingly used in chemical prediction tasks because they can identify complex relationships between molecular structure and biological responses [4]. In this context, molecular information can be transformed into numerical descriptors, allowing supervised learning algorithms to classify compounds according to their expected behavior. However, many high-performing models are difficult to interpret directly [1]. This lack of interpretability is particularly relevant in toxicology, where a prediction is not sufficient by itself [2]. Researchers need to understand which molecular characteristics influenced the model’s decision in order to assess the plausibility of the results, generate new hypotheses, and increase confidence in their use.

Explainable Artificial Intelligence (XAI) is a research field dedicated to developing methods and techniques that make it possible to understand, interpret, and justify the behavior and decisions of artificial intelligence systems, with the aim of improving their transparency, reliability, and auditability [1], [2]. XAI addresses this challenge through feature-attribution techniques that estimate the contribution of each input variable

to the model prediction, with SHAP and LIME being two of the most widely used post-hoc methods [1]. However, they may highlight different features for the same prediction. In addition to generating explanations, it is also necessary to communicate their results clearly and understandably to users [10]. In this regard, Large Language Models (LLMs) offer a promising alternative for translating technical explanations into natural-language descriptions, although their use must preserve fidelity to the evidence provided by XAI methods.

In this context, the objective of this work is to develop and evaluate a workflow for molecular toxicity prediction, explain the resulting models using SHAP and LIME, generate a consensus strategy to combine their explanations, and incorporate an LLM-based communication layer. In particular, this study analyzes whether the consensus strategy produces more robust and consistent explanations than those generated by each method individually, and provides an LLM-mediated narrative layer to enhance the communication of these explanations without altering the evidence provided by the XAI.

II. BACKGROUND

A. Predictive Modeling

In cheminformatics, Quantitative Structure–Activity Relationship (QSAR) models establish quantitative relationships between molecular structures and their associated chemical or biological properties. For this purpose, chemical structures are represented through Molecular descriptors (MDs), which are used as input variables for statistical or ML models. Based on the relationships learned between these descriptors and the property of interest, the model can predict that property for new compounds [3], [4].

B. Model Explanation

Many ML models behave as black boxes because they generate predictions without making the process leading to those predictions directly interpretable. XAI methods seek to make this process more transparent and identify the factors influencing each result. SHAP and LIME are model-agnostic and provide feature-level explanations for predictions [1]. SHAP estimates the contribution of each input feature using Shapley values from cooperative game theory, providing an additive explanation of the prediction [5]. LIME explains an

individual prediction by approximating the behavior of the original model with a locally interpretable surrogate model built around the instance of interest [6]. Because these methods rely on different assumptions, they can produce complementary or divergent explanations for the same prediction.

Different XAI techniques often emphasize different aspects of a model’s decision, even when explaining the same prediction. As a result, explanations obtained from a single method may not fully characterize the model behavior. Comparing multiple explanation methods has therefore been proposed as a way to identify consistent patterns and assess the robustness of feature attributions. Consensus-based approaches seek to identify features that are consistently highlighted across different explanation methods while distinguishing them from method-dependent or uncertain explanations. Rather than assuming that a single explanation is definitive, these approaches exploit agreement and disagreement among explainers to improve the interpretation of model predictions.

C. Large Language Models as a communication layer

Although XAI methods provide valuable information about model behavior, their outputs are often numerical or visualization-based, making them difficult for non-specialists to interpret. LLMs offer the ability to transform structured explanation outputs into coherent natural-language descriptions, facilitating communication between ML models and end users.

Recent studies, approaches such as XpertAI [11], integrate XAI methods with LLMs to generate human-interpretable structure–property explanations supported by external scientific evidence. XpertIA uses generative AI to produce the relationship, not to narrate it. Our proposal highlights the potential of LLMs as interfaces between explainability methods and users, while emphasizing the importance of preserving fidelity between the generated narrative and the evidence provided by ML plus XAI methods. Ensuring that LLM-generated explanations faithfully reflect the underlying XAI outputs remains an active research challenge [9].

III. MATERIALS AND METHODS

The overall workflow of the proposed methodology is illustrated in Fig. 1.

A. Model training and evaluation

The TIRESIA developmental-toxicity dataset was used in this study [7]. This dataset contains molecular structures represented as SMILES strings and a binary toxicity label. The MDs were calculated using RDKit [8]. The dataset was divided into training and testing subsets using a stratified split, preserving the proportion of toxicant and non-toxicant compounds. Several supervised classification algorithms were trained. The models were evaluated using accuracy, precision, recall and F1-score. To continue the XAI analysis, the three models with the highest F1 scores were selected.

B. Explainability, consensus and LLM-based communication

To enable a controlled comparison of local explanation methods, the analysis was conducted on a single representative molecule. The same instance was used for SHAP, LIME, consensus analysis, and LLM explanation generation, ensuring that differences in the explanations reflected the methods themselves rather than variations across molecular instances.

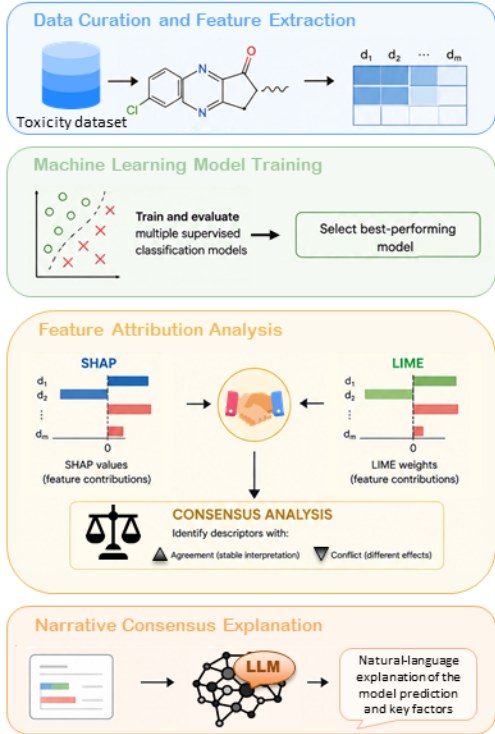


Fig. 1. Overview of the proposed workflow combining predictive modeling, SHAP and LIME explanations, consensus analysis, and LLM-based communication for molecular toxicity prediction.

Since SHAP and LIME rely on different theoretical principles, their explanations may differ in the descriptors identified as relevant, their estimated importance, or the direction of their contributions. To enable a systematic comparison, the outputs were transformed into a common representation containing the descriptor name, relative importance, contribution direction, and ranking position. Each descriptor was assigned to one of three categories according to the level of agreement between both methods, as summarized in Table I.

Descriptors were classified as strong consensus when SHAP and LIME assigned the same contribution sign and their ranking positions differed by no more than one. A ranking difference of two or more positions with the same sign was classified as weak consensus, while opposite contribution signs were classified as conflict. These thresholds correspond to the implementation used in the consensus module.

Finally, a LLM (gemini-2.5-flash) was employed exclusively as a communication layer to transform the structured explainability results into a natural-language explanation. The prompt provided to the LLM included the model prediction, the

TABLE I
CONSENSUS CATEGORIES USED TO COMPARE SHAP AND LIME EXPLANATIONS.

Category	Condition	Interpretation
Strong consensus	Same sign and ranking difference of at most one position	More consistent interpretation
Weak consensus	Same sign and ranking difference of two or more positions	Interpretation with caution
Conflict	Opposite contribution signs, regardless of ranking	Uncertain interpretation

MDs, the SHAP and LIME outputs, the consensus categories, and explicit interpretation rules. The model was instructed to avoid introducing chemical knowledge beyond the available evidence and to explicitly report disagreements between explanation methods when present. Consequently, the LLM did not participate in toxicity prediction, feature attribution, or validation of chemical mechanisms, but only communicated the evidence produced by the ML model and the explainability analysis.

C. Narrative Fidelity and Factuality Evaluation

The generated explanations were evaluated along three complementary dimensions: fidelity, hallucination, and factuality. Let V_{xai} be the consensus descriptors obtained from XAI and V_{llm} those mentioned by the LLM.

Fidelity. Coverage was measured using Jaccard, precision, and recall:

$$J = \frac{|V_{xai} \cap V_{llm}|}{|V_{xai} \cup V_{llm}|}, \quad P = \frac{|V_{xai} \cap V_{llm}|}{|V_{llm}|}, \quad R = \frac{|V_{xai} \cap V_{llm}|}{|V_{xai}|}.$$

Ranking preservation was evaluated with Spearman’s (F_r), pairwise ordering with Kendall’s (F_l), and direction preservation as the proportion of shared descriptors with matching contribution signs (F_d). These measures were combined into a composite fidelity score: $Fid = (F_r + F_l + F_d)/3$.

Additionally, we computed a discrete Narrative Fidelity Score (NFS), where M denotes correctly mentioned descriptors, D those with preserved direction, and C those assigned to the correct consensus category. $NFS = (M + D + C)/3N$,

Hallucination. The hallucination rate measures descriptors introduced by the LLM that are absent from the XAI consensus: $Hall = (|V_{llm} \setminus V_{xai}|)/|V_{llm}|$.

Factuality. Each explanation was decomposed into atomic propositions and automatically verified against the XAI output. Propositions (total N_t) were classified as correct (N_c), partially correct (N_p), or unsupported, yielding the factuality score: $F = (N_c + 0.5N_p)/N_t$.

Together, these metrics distinguish information preservation (fidelity), unsupported additions (hallucination), and correctness of generated claims (factuality).

D. Experimental Workflow

IV. RESULTS AND DISCUSSION

The dataset was divided into stratified training and test subsets containing 187 and 47 molecules, respectively, corresponding approximately to an 80/20 split. Table II reports the classification performance of all evaluated models on the

test set. The three highest-performing models according to F1-score are highlighted in bold. Extra Trees achieved the highest F1-score and was therefore selected to continue the analysis.

TABLE II
CLASSIFICATION PERFORMANCE OF THE EVALUATED MODELS ON THE TEST SET.

Model	Acc.	Prec.	Rec.	F1
Extra Trees	0.787	0.806	0.906	0.853
Random Forest	0.766	0.784	0.906	0.841
Grad. Boosting	0.766	0.800	0.875	0.836
KNN	0.723	0.757	0.875	0.812
SVM	0.681	0.681	1.000	0.810
Naive Bayes	0.702	0.725	0.906	0.806
MLP	0.681	0.718	0.875	0.789
Log. Regression	0.638	0.692	0.844	0.761

The following results correspond to the local explanation generated for the molecule allantoin (SMILES: NC(=O)NC1NC(=O)NC1=O), classified as *Non-Toxicant*. This molecule was selected as the representative case study and was consistently used throughout the local explainability, consensus analysis, and LLM-based narrative evaluation.

For visualization purposes, only the six most relevant MDs are displayed in the local SHAP and LIME explanations (Figs. 3(a) and 3(b)). They correspond to molecular weight (MolWt), octanol–water partition coefficient (LogP), number of hydrogen bond donors (HDonors), number of hydrogen bond acceptors (HAcceptors), topological polar surface area (TPSA), and number of rotatable bonds (RotBonds). Showing only these descriptors does not represent a substantial reduction of the explanation, as $FI_{0.8}$ indicates that three to four descriptors are sufficient to capture at least 80% of the total feature importance across the evaluated models. Specifically, for Extra Trees and Random Forest, four descriptors were required for SHAP and LIME, for Gradient Boosting, SHAP required only three descriptors, while LIME required four.

As shown in Fig. 3, SHAP and LIME attributed relevance to the same six MDs, but they did not produce the same ranking. SHAP assigned the greatest contribution to LogP, followed by HDonors and RotBonds, whereas LIME ranked RotBonds first, followed by LogP and MolWt. This difference does not necessarily indicate contradictory explanations, since SHAP and LIME estimate feature relevance using different principles and scales. Instead, it shows that agreement on the set of relevant descriptors does not imply agreement on their relative order or magnitude. This ranking variability motivated the subsequent consensus analysis.

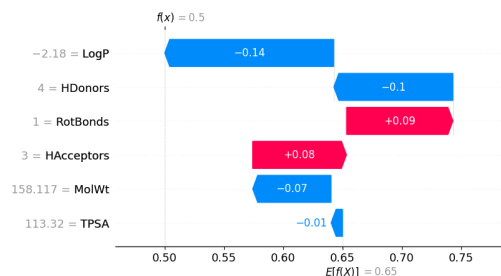


Fig. 3(a). Local SHAP explanation for the selected test molecule using the Extra Trees classifier.

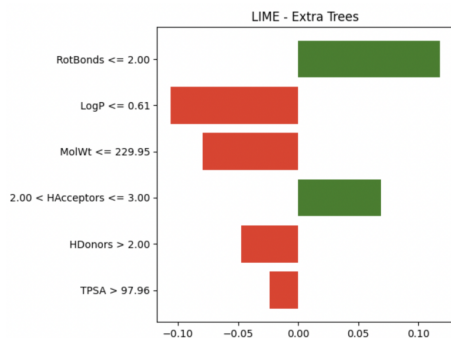


Fig. 3(b). Local LIME explanation for the selected test molecule using the Extra Trees classifier.

As illustrated in Fig. 4, SHAP and LIME identified the same six MDs as relevant but assigned them different ranking orders. SHAP ranked LogP as the most influential descriptor, followed by HDonors and RotBonds, whereas LIME ranked RotBonds first, followed by LogP and MolWt. These differences are expected because SHAP and LIME estimate feature importance using different principles. Therefore, agreement on the relevant descriptors does not necessarily imply agreement on their relative ranking.

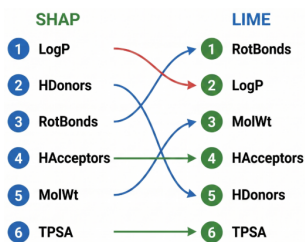


Fig. 4. Comparison of descriptor rankings obtained with SHAP and LIME for the selected molecule. Arrows connect the same MDs across both methods, illustrating that although the same descriptors are identified as relevant, their relative ranking differs.

The explanations produced by the LLM followed a structured template that summarized the agreement and discrepancies between SHAP and LIME, highlighted the most influential MDs, and provided guidance on the reliability of the interpretation. Owing to space limitations, a representative excerpt of the generated narrative is presented:

Toxicity Prediction Explanation for Molecule 1

1. Overall model results

For Molecule 1, the models produced different predictions:

- The **Extra Trees** model predicts the molecule as **Non-Toxicant**, with a probability of...

2. Most important variables according to the SHAP-LIME consensus

Extra Trees (Prediction: Non-Toxicant)

- **LogP** (a measure of lipophilicity) was the most influential descriptor and showed a **negative effect**, strongly pushing the prediction toward the **Non-Toxicant** class.
- **HDonors** (hydrogen bond donors) and **MolWt** (molecular weight) also contributed toward the **Non-Toxicant** prediction, although their importance rankings differed between SHAP and LIME.
- In contrast, **RotBonds** (molecular flexibility) and **HAcceptors** (hydrogen bond acceptors) showed a **positive effect**, pushing the prediction toward the **Toxicant** class, although with a smaller overall influence.

3. Cautious chemical and toxicological interpretation

4. Conflicts and warnings between SHAP and LIME

5. Final conclusion

Molecule 1 was predominantly classified as **Toxicant**, particularly by the Gradient Boosting model. The main descriptors driving this prediction were **RotBonds** and **HAcceptors**, whereas **LogP**, **MolWt**, and **HDonors** generally pushed the prediction toward the **Non-Toxicant** class.

Nevertheless, the conflicts identified for **LogP** and **TPSA** in the Gradient Boosting model indicate that these descriptors should be interpreted with caution.

A. Fidelity, robustness and consensus consistency evaluation

The explanations were evaluated using fidelity, robustness, consensus consistency, and Rank Alignment (see Table III). LIME fidelity measured how closely the local surrogate reproduced the original classifier around the selected molecule, whereas SHAP fidelity was assessed by reconstructing the model output from the SHAP contributions. Robustness measured the stability of the explanations under small perturbations of the MDs.

TABLE III
EVALUATION METRICS FOR SHAP, LIME AND LLM EXPLANATIONS.

Model	Fid. LIME	Rob. LIME	Fid. SHAP	Rob. SHAP	Agree. (%)	Cons.
Extra Trees	0.383	0.946	1.000	0.997	100.00	1.000
Random Forest	0.385	0.955	1.000	0.981	100.00	1.000
Gradient Boosting	0.301	0.985	1.000	0.927	66.67	1.000

Table III shows moderate LIME fidelity and high robustness for all three models. SHAP achieved reconstruction fidelity of 1.000 in every case, indicating exact reconstruction of the model output rather than perfect explanation quality. SHAP-LIME agreement reached 100.00% for Extra Trees and Random Forest and 66.67% for Gradient Boosting, while consensus consistency was 1.000 for all models.

Also, Rank Alignment was calculated using Jaccard similarity over the top-50% most relevant descriptors ($\alpha = 0.5$, three of six descriptors). The scores were 0.500 for Extra Trees and Random Forest and 0.200 for Gradient Boosting, indicating lower overlap between the highest-ranked SHAP and LIME descriptors for Gradient Boosting.

B. Narrative Fidelity Evaluation

The metrics defined in Section III-C were computed for the Extra Trees narrative. The consensus module returned $N =$

$|V_{xai}| = 6$ descriptors, none of them classified as conflict: LogP, HAcceptors and TPSA reached strong consensus, while HDonors, RotBonds and MolWt reached weak consensus. The narrative mentioned $|V_{llm}| = 5$ of them, omitting TPSA, so $|V_{xai} \cap V_{llm}| = 5$ and $|V_{xai} \cup V_{llm}| = 6$.

Coverage was therefore $J = 5/6 = 0.833$, $P = 5/5 = 1.000$ and $R = 5/6 = 0.833$, with a hallucination rate of $Hall = 0/5 = 0.000$. The LLM introduced no descriptor absent from the consensus; the loss of coverage is due exclusively to omission, and the omitted descriptor (TPSA) is the one with the smallest attributed magnitude in both explainers.

Directional fidelity was complete, $F_d = 5/5 = 1.000$, since the five mentioned descriptors preserved their contribution sign. Ranking fidelity was assessed by deriving the LLM ranking from the order and grouping of the narrative (LogP first; HDonors and MolWt tied; RotBonds and HAcceptors tied) and comparing it with the mean of the SHAP and LIME positions, yielding $F_r = 0.433$ and $F_l = 0.354$. The composite fidelity score was 0.596

Ranking is thus the weakest dimension. Computed against each explainer separately, the narrative ordering correlates with SHAP ($\rho = 0.527$) but not with LIME ($\rho = -0.053$), indicating that the generated text implicitly reproduced the SHAP ordering rather than a balanced consensus ordering.

For the discrete score, $M = 5$ descriptors were correctly mentioned and $D = 5$ preserved their direction. The consensus category was correctly represented for $C = 4$: the narrative reported the ranking disagreement for HDonors and MolWt and presented LogP and HAcceptors without reservation, consistent with their strong consensus, but grouped RotBonds with HAcceptors and omitted its weak-consensus status. Hence NFS reaches 0.778. The narrative preserved direction fully, coverage almost fully, and the reliability categories partially.

Factuality was assessed over $N_t = 10$ statements, of which eight were correct, one partially correct and one non-factual, giving $Fac = (8 + 0.5)/10 = 0.850$. The non-factual statement is the class assignment in the final section of the narrative, which reports the molecule as predominantly Toxicant although the Extra Trees prediction is Non-Toxicant. Taken together, $P = 1.000$ and $Hall = 0.000$ with $NFS < 1$ and $Fac < 1$ show that the gap does not arise from invented descriptors but from omitted evidence and from unsupported claims about descriptors that are themselves real.

V. LIMITATIONS AND CONCLUSIONS

This work has some limitations. The molecular representation was restricted to six physicochemical descriptors for the XAI analysis, which may not capture all structural factors associated with toxicity. For brevity, the explainability analysis is presented for a representative test molecule, although similar qualitative trends were observed across the remaining analyzed molecules. Furthermore, SHAP and LIME explain model behavior rather than causal toxicological mechanisms, and the LLM was used exclusively as a communication layer rather than as a source of scientific validation.

This work presented an explainable workflow for molecular toxicity prediction that combines supervised machine learning, complementary XAI methods, a consensus analysis, and an LLM-based communication layer. Among the evaluated classifiers, Extra Trees achieved the highest predictive performance and was selected together with the two next best-performing models for explainability analysis. The comparison between SHAP and LIME allowed consistent and divergent feature attributions to be explicitly identified, providing structured evidence for interpretation.

The proposed evaluation framework further showed that LLM-generated narratives can preserve most of the information provided by the XAI methods while still introducing omissions, ranking inconsistencies, or unsupported claims. These results demonstrate that descriptor coverage alone is insufficient to assess explanation quality and highlight the need to evaluate fidelity, factuality, and hallucinations when LLMs are used to communicate XAI outputs. Overall, the proposed workflow provides a structured approach for generating and assessing natural-language explanations while preserving the distinction between model evidence and generated narratives.

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